

Systems of Equations

A system of equations is a collection of two or more equations that use the same variables. The goal is to find the value of each variable that makes all equations true at the same time. In algebra, one of the most common examples is a system of two linear equations with two variables, usually written using x and y . A solution to the system is an ordered pair (x, y) that satisfies every equation in the system.

For example, a simple system can be written as:

$$\begin{cases} x - y = 5 \\ 2x + y = 7 \end{cases}$$

This system asks us to find one value of x and one value of y that make both equations true. Systems of equations are useful because many real-world problems involve more than one condition. For example, in engineering, economics, physics, and data science, we often need to solve several relationships at the same time.